

Data analysis: Airbnb listings price in Seattle, WA

- The data set we looked at collects data describing listings activities of homestays from Airbnb in Seattle, WA.
- We chose the following predictors and response variable for the initial model
- The predictor variables that we highlighted are the primary variables we will be using to model with. That is, we will be utilizing priors for these variables and focusing on creating a linear regression model with just the four highlighted variables. The other predictor variables might be used later in our analysis with default priors.

1. Predictors

Quantitative variables:

host_response_rate: the rate of host's responses

host_total_listings_count: The total number of listings managed by the host across all Airbnb platforms

accommodates: number of people the property can accommodate for sleeping arrangements

bathrooms: the total number of bathrooms in the property

bedrooms: number of bedrooms in the property

beds: number of beds in the property

extra_people: the additional charge per person for guests beyond the number of guests included in the base price

minimum_nights: minimum number of nights one has to book

number_of_reviews: number of reviews a listing has received from guests over its lifetime

review_scores_rating: overall rating given to a listing by guests, based on their reviews

reviews_per_month: the average number of reviews a listing receives per month

- *Qualitative variables:*

host_identity_verified: Indicates whether the host's identity has been verified by Airbnb

property_type: the type of property

room_type: the type of room guests will be staying in

bed_type: the type of bed available in the listing

instant_bookable: whether the listing can be booked immediately without the host's approval

cancellation_policy: the host's cancellation policy, which dictates how flexible guests are regarding cancellations

2. Response variable

- **price:** price of the property on Airbnb

Our Models

Main Effects, First Order:

- $\text{Price} = -44.1 + 12.6(\text{cancellation_policy_moderate}) + 24.7(\text{cancellation_policy_strict}) + 43.5(\text{beds}) + 0.855(\text{review_scores_rating})$

Alternative 1:

- $\text{Price} = 36.5 + 4.25(\text{beds}^2) + 7.98(\text{bathrooms}^2) + 0.596(\text{review_scores_rating})$

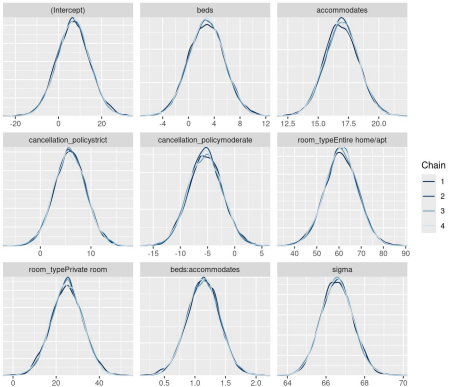
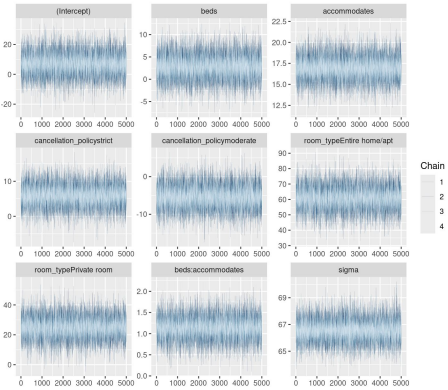
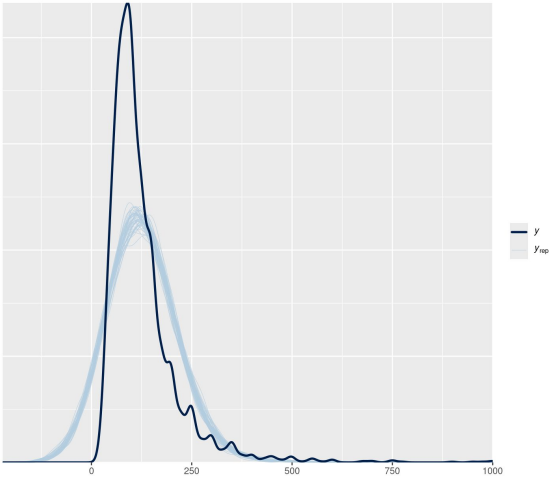
Alternative 2:

- $\text{Price} = 6.87 + 2.82(\text{beds}) + 16.9(\text{accommodates}) + 5.85(\text{cancellation_policy_strict}) - 5.40(\text{cancellation_policy_moderate}) + 61(\text{room_typeEntire home/apt}) + 25(\text{room_typePrivate room}) + 1.14(\text{beds} * \text{accommodates})$

Default Priors:

- $\text{Price} = -33.7 + 43.6(\text{beds}) + .809(\text{review_scores_rating}) + 1.34(\text{cancellation_policy_moderate}) + 19.2(\text{cancellation_policy_strict})$

Best model: Price = 6.87 + 2.82(beds) +16.9(accommodates) + 5.85(cancellation_policy_strict) - 5.40(cancellation_policy_moderate) + 61(room_typeEntire home/apt) + 25(room_typePrivate room) + 1.14(beds*accommodates)



mae	mae_scaled	within_50	within_95
25.85055	0.3880156	0.7207977	0.9639126
mae	mae_scaled	within_50	within_95
26.10535	0.3920606	0.7160589	0.9629546

Posterior credible interval:

term	estimate	std.error	conf.low	conf.high
<chr>				
<dbl> <dbl> <dbl> <dbl>				
1 (Intercept)	6.87	7.52	-5.62	19.4
2 beds	2.82	2.75	-1.70	7.34
3 accommodates	16.9	1.40	14.7	19.3
4 cancellation_policystrict	5.85	3.14	0.736	11.0
5 cancellation_policymoderate	-5.40	3.13	-10.5	-0.351
6 room_typeEntire home/apt	61.0	7.34	48.6	73.2
7 room_typePrivate room	25.0	7.35	12.8	37.1
8 beds:accommodates	1.14	0.265	0.689	1.58
9 sigma	66.6	0.837	65.2	68.0
10 mean_PPD	127.	1.67	124.	130.